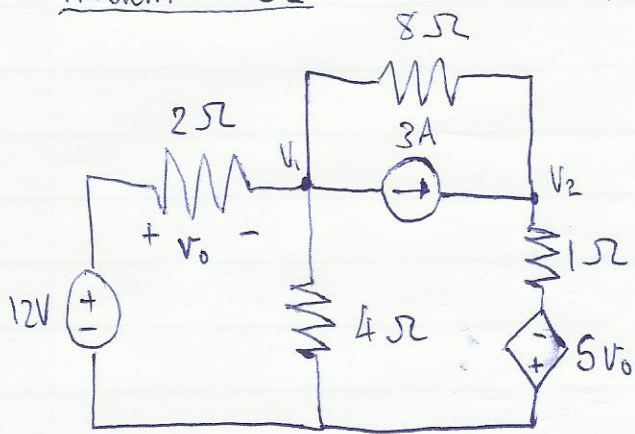


Problem 3.22



Nodal analysis:

Determine V_1 and V_2

Note 1: $\frac{V_1 - 12}{2} + \frac{V_1 - V_2}{8} + 3 + \frac{V_1}{4} = 0$

$$\Rightarrow 7V_1 - V_2 = 24 \quad \dots \textcircled{1}$$

Node 2: $\frac{V_2 - V_1}{8} - 3 + \frac{V_2 - (-5V_0)}{1} = 0$

but $V_0 = 12 - V_1$

$$\Rightarrow -41 V_1 + 9 V_2 = -456 \quad \dots (2)$$

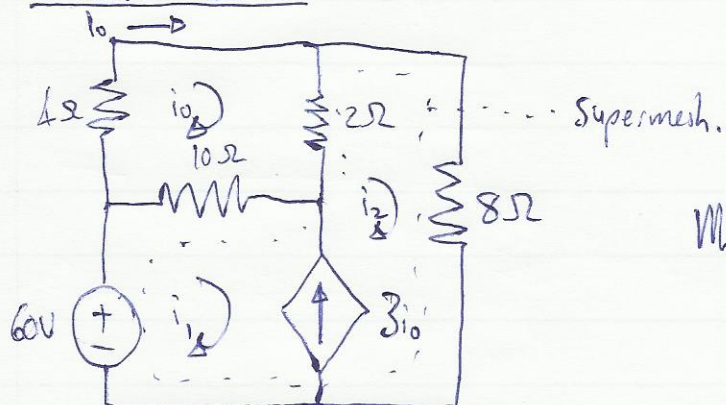
solve (1) and (2) simultaneously:

$$V_1 = -10,9 \text{ V}$$

$$V_2 = -100,4 \text{ V}$$



Problem 3.50



Mesh analysis:
Determine i_o

Mesh 0:

$$4i_o + 2(i_o - i_2) + 10(i_o - i_1) = 0$$

$$\Rightarrow 16i_o - 10i_1 - 2i_2 = 0 \quad \dots \textcircled{1}$$

Mesh 1 & 2:

$$-60 + 10(i_1 - i_o) + 2(i_2 - i_o) + 8i_2 = 0$$

$$\Rightarrow -12i_o + 10i_1 + 10i_2 = 60 \quad \dots \textcircled{2}$$

Current source:

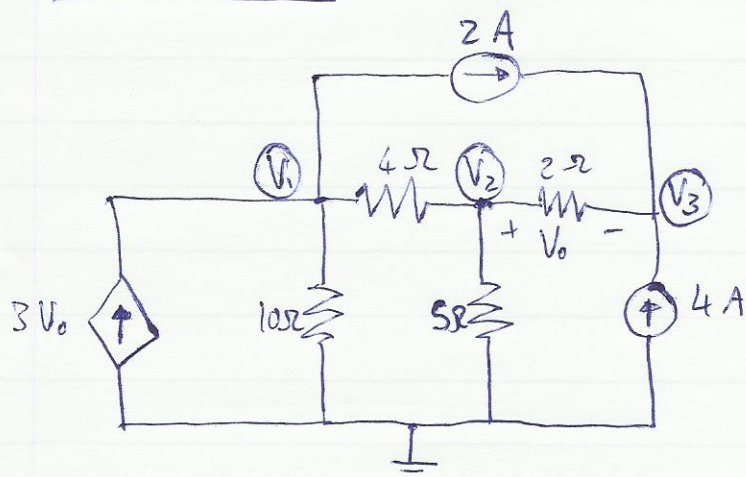
$$3i_o = -i_1 + i_2$$

$$\Rightarrow -3i_o - i_1 + i_2 = 0 \quad \dots \textcircled{3}$$

Solve $\textcircled{1}$; $\textcircled{2}$; $\textcircled{3}$ simultaneously:

$$\underline{i_o = 1.73 \text{ A} \rightarrow}$$

Problem 3.67



Nodal, by inspection.
Only matrices.

KCL @ node 1: (currents out of node positive)

$$-3V_0 + 2 + \frac{V_1 - V_2}{4} + \frac{V_1}{10} = 0 \quad \text{but } V_0 = V_2 - V_3$$

$$\Rightarrow 7V_1 - 65V_2 + 60V_3 = -40 \quad \dots \textcircled{1}$$

KCL @ node 2:

$$-\frac{V_2 - V_1}{4} + \frac{V_2}{5} + \frac{V_2 - V_3}{2} = 0$$

$$\Rightarrow -5V_1 + 19V_2 - 10V_3 = 0 \quad \dots \textcircled{2}$$

KCL @ node 3:

$$-2 - 4 + \frac{V_3 - V_2}{2} = 0$$

$$\Rightarrow -V_2 + V_3 = 12$$

Resulting matrix:

$$\begin{bmatrix} 7 & -65 & 60 \\ -5 & 19 & -10 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} -40 \\ 0 \\ 12 \end{bmatrix}$$

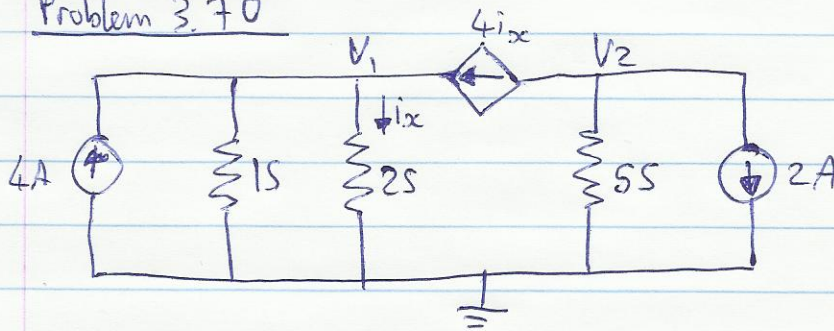
Solution: (not required)

$$V_1 = -164.21 \text{ V}$$

$$V_2 = -77.90 \text{ V}$$

$$V_3 = -65.90 \text{ V}$$

Problem 3.70



Nodal, by inspection.
Only matrices.

Node 1:

$$\begin{aligned} -4 + (2+1)V_1 - 4i_x &= 0 \\ \Rightarrow 3V_1 - 4i_x &= 4 \quad \dots (1) \end{aligned}$$

Node 2:

$$\begin{aligned} 4i_x + 5V_2 + 2 &= 0 \\ \Rightarrow 5V_2 + 4i_x &= -2 \quad \dots (2) \end{aligned}$$

current controlled current source:

$$\begin{aligned} i_x &= 2V_1 \\ \Rightarrow 2V_1 - i_x &= 0 \quad \dots (3) \end{aligned}$$

Resulting matrix:

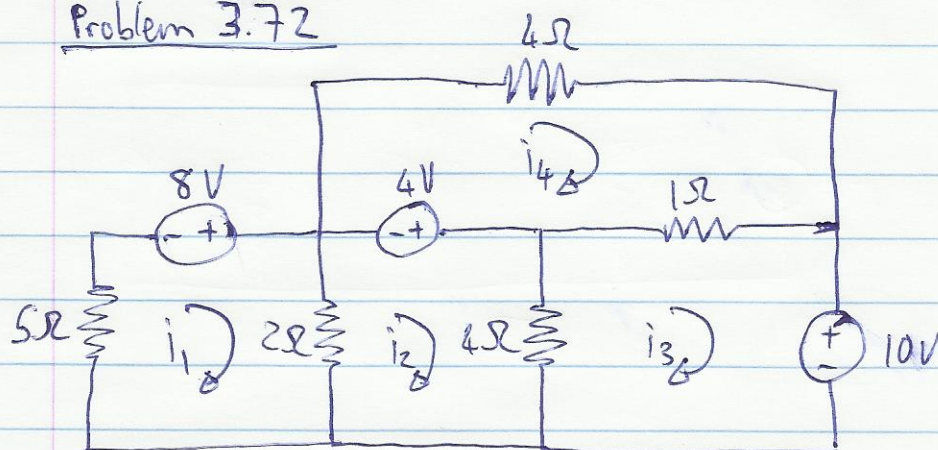
solutions: (not required)

$$\begin{bmatrix} 3 & 0 & -4 \\ 0 & 5 & 4 \\ 2 & 0 & -1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ i_x \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \\ 0 \end{bmatrix}$$

$$V_1 = -0,8V$$

$$V_2 = 0,88V$$

Problem 3.72



Mesh, by inspection only matrices.

$$\text{Mesh 1: } (5+2)i_1 - 2i_2 - 8 = 0$$

$$\Rightarrow 7i_1 - 2i_2 = 8$$

$$\text{Mesh 2: } (2+4)i_2 - 2i_1 - 4i_3 - 4 = 0$$

$$\Rightarrow -2i_1 + 6i_2 - 4i_3 = 4$$

$$\text{Mesh 3: } (4+1)i_3 - 4i_2 - 1i_4 + 10 = 0$$

$$\Rightarrow -4i_2 + 5i_3 - i_4 = -10$$

$$\text{Mesh 4: } (4+1)i_4 - 1i_3 + 4 = 0$$

$$\Rightarrow -i_3 + 5i_4 = -4$$

Matrix:

$$\begin{bmatrix} 7 & -2 & 0 & 0 \\ -2 & 6 & -4 & 0 \\ 0 & -4 & 5 & -1 \\ 0 & 0 & -1 & 5 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \\ -10 \\ -4 \end{bmatrix}$$

Solutions: (not required)

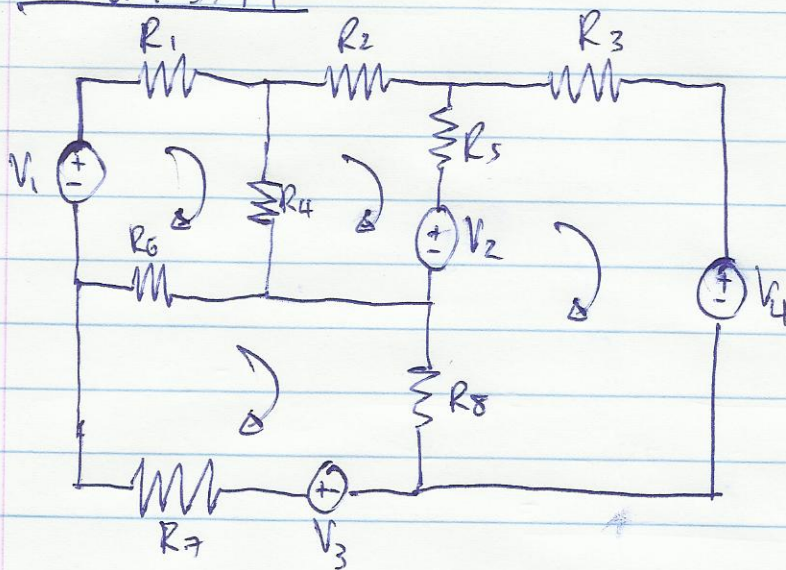
$$i_1 = 0.777A$$

$$i_2 = -1.30A$$

$$i_3 = -3.33A$$

$$i_4 = -1.47A$$

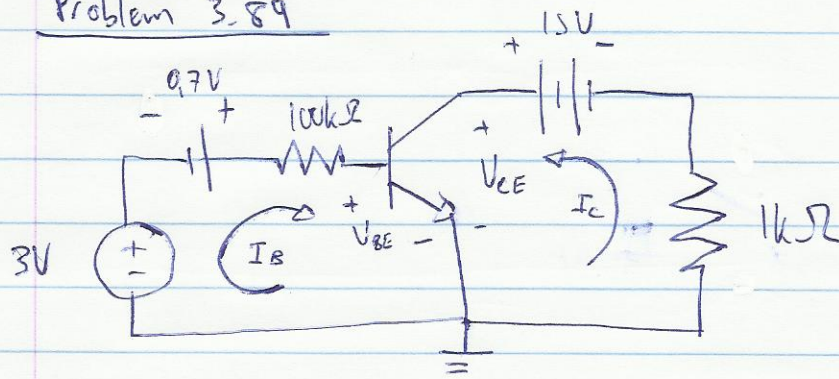
Problem 3.74



Mesh, by inspection.
Only matrices.

$$\begin{bmatrix} R_1 + R_4 + R_6 & -R_4 & -R_6 & 0 \\ 0 & R_2 + R_4 + R_5 & 0 & -R_5 \\ -R_6 & 0 & R_6 + R_7 + R_8 & -R_8 \\ 0 & -R_5 & -R_8 & R_3 + R_5 + R_8 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} V_1 \\ -V_2 \\ V_3 \\ -V_4 \end{bmatrix}$$

Problem 3.89



$$V_{BE} = 0.7 \text{ V}$$

$$\beta = 100$$

Determine I_B and V_{CE}

KCL on I_B loop:

$$-3 - 0.7 + I_B(100k) + V_{BE} = 0$$

$$V_{BE} = 0.7$$

$$\therefore I_B(100k) = 3$$

$$\underline{I_B = 30 \mu\text{A}}$$

$$\begin{aligned} I_C &= \beta \cdot I_B \\ &= 100(30 \mu) \\ &= 3 \text{ mA} \end{aligned}$$

KCL on I_C loop:

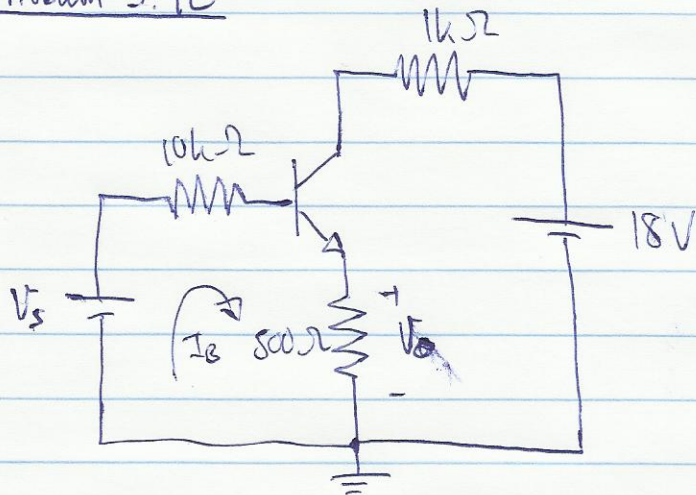
$$1k(I_C) - 15 + V_{CE} = 0$$

$$V_{CE} = 15 - (1k)(3m)$$

$$= 15 - 3$$

$$\underline{= 12 \text{ V}}$$

Problem 3.90



$$V_o = 4V$$

$$\beta = 150$$

$$V_{BE} = 0.7V$$

Determine V_s .

I_B loop:

$$-V_s + I_B(10k) + V_{BE} + I_E(500) = 0$$

$$\text{but } I_E = (\beta + 1)I_B = 151I_B$$

$$\therefore -V_s + 10kI_B + 0.7 + (151)(500)I_B = 0$$

$$V_s - 10kI_B - 75.5kI_B = 0.7$$

$$V_s - 85.5kI_B = 0.7 \quad \dots (1)$$

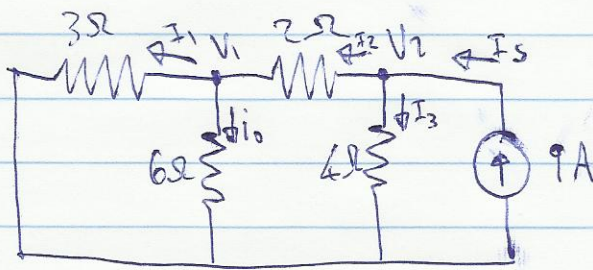
$$\text{But } V_o = 4V \text{ (given)}$$

$$\therefore 75.5kI_B = 4 \quad \dots (2)$$

Solving (1) and (2) simultaneously:

$$\underline{V_s = 5.23V} \quad \square$$

Problem 4.4



Use linearity to determine i_0 .

$$\text{if } i_0 = 1A \Rightarrow V_1 = 6V$$
$$\therefore I_1 = \frac{6V}{3\Omega} = 2A$$

$$I_2 = I_1 + i_0 = 2 + 1 = 3A$$

$$V_2 = V_1 + I_2(2\Omega)$$
$$= 6 + 3(2) = 12V$$

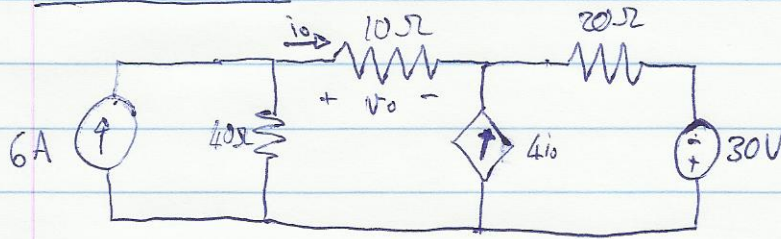
$$\therefore I_3 = \frac{12V}{4\Omega} = 3A$$

$$I_s = I_2 + I_3 = 3 + 3 = 6A$$

$$I_s = 6A \rightarrow i_0 = 1A$$

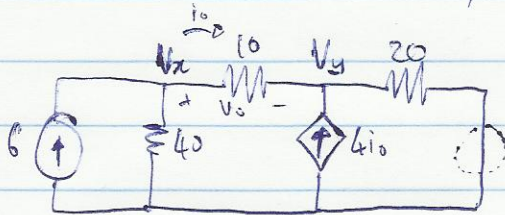
$$\therefore I_s = 9A \rightarrow \underline{i_0 = 1.5A}$$

Problem 4.11



Use superposition to find v_o and i_o .

Consider current source first, voltage source $\rightarrow$ short circuit.



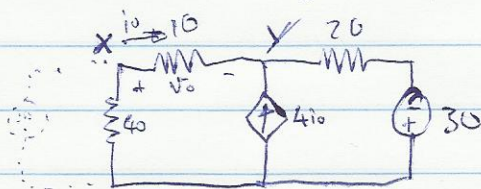
$$\begin{aligned} \text{at node } x: \frac{V_x - V_y}{10} + \frac{V_x}{40} - 6 &= 0 \\ \Rightarrow 5V_x - 4V_y &= 240 \dots (1) \end{aligned}$$

$$\begin{aligned} \text{at node } y: \frac{V_y - V_x}{10} + \frac{V_y}{20} - 4i_o &= 0 \quad \text{but } i_o = \frac{V_x - V_y}{10} \\ \Rightarrow -10V_x + 11V_y &= 0 \dots (2) \end{aligned}$$

solving (1) and (2):

$$\begin{aligned} V_x &= 176V & \therefore V_o &= V_x - V_y = 16V \\ V_y &= 160V & i_o &= \frac{V_o}{10} = 1.6A \end{aligned}$$

Now consider voltage source, current source $\rightarrow$ open circuit



$$\begin{aligned} \text{at node } x: \frac{V_x - V_y}{40} + \frac{V_x}{40} &= 0 \\ \Rightarrow 5V_x - 4V_y &= 0 \dots (3) \end{aligned}$$

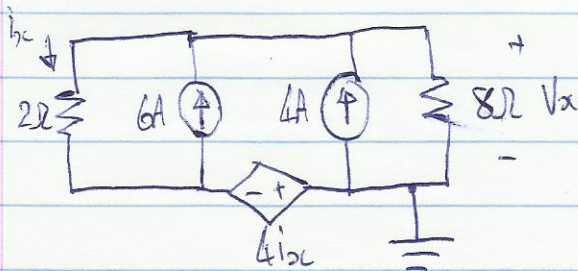
solving (3) and (4):

$$\begin{aligned} V_x &= -8V \\ V_y &= -10V \\ \therefore V_o &= 2V \\ i_o &= 0.2A \end{aligned}$$

$$\begin{aligned} \text{at node } y: \frac{V_y - (-30)}{20} + \frac{V_y - V_x}{10} - 4i_o &= 0 \\ \Rightarrow -10V_x + 11V_y &= -30 \dots (4) \end{aligned}$$

$$\begin{aligned} \text{Final result: } V_o &= 16 + 2 = 18V \\ i_o &= 1.6 + 0.2 = 1.8A \end{aligned}$$

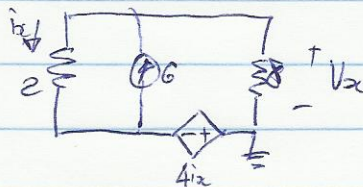
Problem 4.19



Determine V_x
using superposition.

Note selection of $\underline{\underline{9}}$!

Considering 6A current source first:



$$i_x - 6 + \frac{V_x}{8} = 0$$

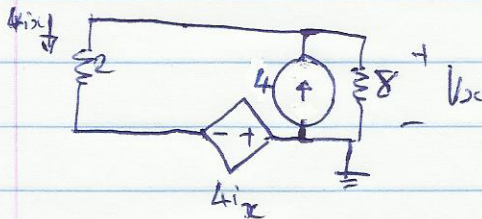
$$i_x = \frac{V_x - (-4i_x)}{2}$$

$$\Rightarrow i_x = -\frac{1}{2} V_x$$

$$\text{thus } -\frac{1}{2} V_x - 6 + \frac{1}{8} V_x = 0$$

$$\Rightarrow V_x = -16 \text{ V}$$

consider now 4A current source:



$$i_x - 4 + \frac{V_x}{8} = 0$$

$$\text{but } i_x = -\frac{1}{2} V_x$$

$$\Rightarrow V_x = -10,67 \text{ V}$$

Combining results:

$$V_x = (-16) + (-10,67)$$

$$= \underline{\underline{-26,67 \text{ V}}}$$